

claude-windows / a9fd4426-f234-4e4a-bfb4-1f2c8255c661

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-muneem\`a9fd4426-f234-4e4a-bfb4-1f2c8255c661\subagents\workflows\wf_076beba8-59c\agent-aa98012ce24a1bb8d.jsonl`  
- SHA-256: `0100248c6226d5ec673ffed19dd86d02a6fad6291d2a496140a8e670c9dac44f`  
- Source modified: `2026-05-31T04:22:01+00:00`  
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- project: `wf_076beba8-59c`  
- session_id: `a9fd4426-f234-4e4a-bfb4-1f2c8255c661`
```

Transcript

1. user (2026-05-31T04:21:06.023Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Identify the best LOCAL (self-hosted, single NVIDIA CUDA GPU) vision / document-AI / OCR models ? and, as a clearly-flagged approval-gated fallback only, paid cloud API services ? for parsing BANK STATEMENT PDFs into structured transaction tables (date, narration/description, withdrawal/debit, deposit/credit, running balance), for a LOCAL-FIRST personal-finance tool ("muneem").

Target documents: Indian bank statements (HDFC, Axis, ICICI, AU Small Finance) ? dense, multi-column, often SEMI-RULED or borderless tables, sometimes multiple accounts per statement; both born-digital (text layer present) and the harder cases where pdfplumber's extract_tables/coordinate clustering is unreliable (e.g. Axis, whose column layout varies between statements).

HARD CONSTRAINTS (a model that violates these is unusable for us):

1. MUST run 100% LOCALLY for statement contents (highly sensitive). NO cloud APIs by default. Paid/cloud is acceptable ONLY as an explicitly-approved fallback ? still report the best cloud options but label them clearly as "cloud/sensitive ? approval-gated, not default".
2. MUST be usable from Python (transformers / vLLM / Ollama / ONNX Runtime / native lib).
3. License MUST be permissive & commercially safe for BOTH the CODE and the MODEL WEIGHTS: MIT, BSD, Apache-2.0, ISC, HPND. AVOID and explicitly FLAG anything GPL/LGPL/AGPL, CC-BY-NC / non-commercial, research-only, or custom "community"/acceptable-use licenses with restrictions. This is critical ? many document-AI models and their training datasets are non-commercial (e.g. LayoutLMv3 weights are CC-BY-NC; Surya and Marker use GPL-or-commercial dual licensing; several VLM weights ship custom restrictive licenses). Verify the WEIGHTS license, not just the repo license.
4. Hardware: a single consumer/prosumer NVIDIA GPU. Give VRAM tiers (~8-12 GB consumer, ~16-24 GB) and which models/quantizations fit each.

Evaluate and compare these LOCAL approaches ? for each give: code license, MODEL-WEIGHTS license (state explicitly), latest release / maintenance status as of 2025-2026, VRAM need + quantization options, Python integration path, and reliability specifically on DENSE FINANCIAL TABLES / multi-column statements:

- Open vision-language models (VLMs) for document/table understanding: Qwen2-VL and Qwen2.5-VL (note per-size license ? which sizes are Apache-2.0 vs restricted), MiniCPM-V 2.6, InternVL2 / 2.5, Microsoft Phi-3.5-vision, Mistral Pixtral, Meta Llama-3.2-Vision (flag the Llama Community License restrictions), Moondream2, GOT-OCR2.0, olmOCR and other OCR-specialized VLMs.
- OCR-specialist / document-AI pipelines: PaddleOCR + PP-Structure (table recognition), docTR, EasyOCR, Tesseract (already used), Microsoft Table Transformer / TATR (check code AND PubTables-1M dataset/weights license), Donut, Surya (FLAG license), Marker (FLAG license),

MinerU / PDF-Extract-Kit, RapidOCR.

- Table-structure recognition specifically (cell/grid reconstruction for borderless tables):
TATR, PP-Structure, UniTable, etc.

Also assess: for SEMI-RULED / borderless bank tables, is a prompt-driven VLM (page image -> JSON transactions) more reliable than a classic OCR + table-structure pipeline? What are the accuracy/hallucination risks of VLMs on financial NUMBERS (transposed digits, dropped/duplicated rows), and how should output be VERIFIED? (muneem already machine-checks parses by balance reconciliation ? opening + sum(deposits) - sum(withdrawals) == closing, plus per-row running-balance walks and cross-checks against the bank's printed summary ? so any model's output can be automatically validated, which should inform the recommendation.)

Finally, briefly cover the best PAID/CLOUD fallbacks (clearly flagged approval-gated for sensitive data): AWS Textract (AnalyzeExpense / Tables), Google Document AI, Azure AI Document Intelligence, Reducto, LlamaParse, Anthropic/OpenAI vision ? note any self-hosting option and data-handling terms.

Deliver a RANKED shortlist of the most promising LOCAL options for muneem (NVIDIA GPU, Python, commercial-license-safe), each with license (code + weights), VRAM tier, maturity, strengths/weaknesses for dense bank-statement tables, and a concrete recommendation: which local model(s) to prototype for muneem's statement parsing and how they'd combine with the existing deterministic parsers + balance-reconciliation verification (e.g. VLM as a fallback/cross-check when deterministic parsing fails to reconcile). Cite sources and call out any license uncertainty explicitly.

Claim under review

"olmOCR 2 is built on Qwen2.5-VL-7B-Instruct, a 7B-parameter vision-language model, fine-tuned then RL-trained; this anchors its VRAM/hardware footprint to the 7B Qwen2.5-VL tier (roughly 16-24 GB GPU class at FP16, less when quantized)."

****Source:**** <https://arxiv.org/html/2510.19817v1> (primary)

****Supporting quote:**** "We start with a Qwen2.5-VL-7B-Instruct model that has been fine-tuned on olmOCR-mix-1025."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-05-31T04:21:09.755Z)

I'll verify this claim about olmOCR 2's architecture and VRAM footprint. Let me start by loading the web tools and examining the primary source.

3. assistant (2026-05-31T04:21:13.747Z)

I'll fetch the primary source (the olmOCR 2 arXiv paper) to verify the architecture claim.

4. user (2026-05-31T04:21:20.151Z)

olmOCR 2 Base Model and Training Details

Base Model

olmOCR 2 is built on Qwen2.5-VL-7B-Instruct. The paper states: "olmOCR 2 is powered by olmOCR-2-7B-1025, an OCR-specialized VLM trained using reinforcement learning with verifiable rewards (RLVR)."

The model begins with "a Qwen2.5-VL-7B-Instruct model that has been fine-tuned on olmOCR-mix-1025."

Training Approach

The training pipeline involves two main stages:

1. **Supervised Fine-tuning**: "A single epoch of SFT training on olmOCR-mix-1025"
2. **Reinforcement Learning**: "A single epoch of RL training over our synthetic data olmOCR2-synthmix-1025" using "Group Relative Policy Optimization (GRPO)" with binary unit tests as rewards.
3. **Model Averaging**: "Repeating the RL training for six random seeds and averaging (or 'souping') the checkpoints."

Hardware Requirements

The document mentions training used "an 8xH100 GPU node" but provides **no specific information** about inference hardware, VRAM, or computational requirements for running olmOCR 2.

5. assistant (2026-05-31T04:21:23.511Z)

The primary source confirms the core architectural claims. Now let me verify the VRAM/hardware footprint claim, since the paper doesn't specify inference hardware. The claim asserts "roughly 16-24 GB GPU class at FP16, less when quantized" for a 7B Qwen2.5-VL model. Let me check independent sources.

6. user (2026-05-31T04:21:29.952Z)

Web search results for query: "olmOCR 2 7B VRAM GPU requirements inference Qwen2.5-VL-7B"

Links: [{"title": "org/qwen2.5-1m:7b", "url": "https://ollama.com/org/qwen2.5-1m:7b"}, {"title": "Qwen/Qwen2.5-VL-7B-Instruct · Hardware and vram requiremnt to run this model?", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct/discussions/18"}, {"title": "Qwen 2.5 7B VRAM Tips Every Dev Should Know | by Novita AI | Medium", "url": "https://medium.com/@marketing_novita.ai/qwen-2-5-7b-vram-tips-every-dev-should-know-932303373ff0"}, {"title": "Qwen 2.5 7B: Run on 8GB VRAM (Benchmarks + Ollama Pull Command) | Local AI Master", "url": "https://localaimaster.com/models/qwen-2-5-7b"}, {"title": "Qwen2.5-7B: Specifications and GPU VRAM Requirements", "url": "https://apxml.com/models/qwen2-5-7b"}, {"title": "Qwen2.5 VL 72B VRAM Needs: Why Vision Models ... - Novita", "url": "https://blogs.novita.ai/qwen2-5-vl-72b-vram/"}, {"title": "Qwen/Qwen2.5-VL-3B-Instruct · How much VRAM is required? I have 8gb RTX 3060 and seems in sufficient", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-3B-Instruct/discussions/20"}, {"title": "Qwen2-7B: Specifications and GPU VRAM Requirements", "url": "https://apxml.com/models/qwen2-7b"}, {"title": "org/qwen2.5-1m:14b", "url": "https://ollama.com/org/qwen2.5-1m:14b"}]

Based on the search results, here's what I found regarding GPU VRAM requirements for the models mentioned in your query:

Qwen2.5-VL-7B

A 24GB VRAM GPU like the RTX 4090 can run Qwen2.5-VL-7B without issues with flash_attention_2. However, some users report memory challenges even with high-end GPUs when using vLLM.

Qwen2.5-7B (Text Model)

For the text-only version:

- FP16 inference for Qwen 2.5-7B requires ~17 GB VRAM, while FP32 needs over 32 GB
- Quantization (8-bit or 4-bit) allows models to run on smaller GPUs (e.g., RTX 3060 12GB), but with trade-offs in precision

General Recommendations

Additional VRAM is used for the model's activation memory (especially at long context lengths) and transient buffers, and a buffer of ~20% extra VRAM is recommended for safe inference.

Note: I could not find specific information about "olmoCR 2" in the search results, so I cannot provide details about that particular model's requirements.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. user (2026-05-31T04:21:29.953Z)

Web search results for query: "Qwen2.5-VL-7B-Instruct VRAM requirement FP16 inference GB"

Links: [{"title": "Qwen 2.5 7B VRAM Tips Every Dev Should Know | by Novita AI | Medium", "url": "https://medium.com/@marketing_novita.ai/qwen-2-5-7b-vram-tips-every-dev-should-know-932303373ff0"}, {"title": "Qwen/Qwen2.5-VL-7B-Instruct · Hardware and vram requiremnt to run this model?", "url": "https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct/discussions/18"}, {"title": "Mungert/Qwen2.5-VL-7B-Instruct-GGUF · Hugging Face", "url": "https://huggingface.co/Mungert/Qwen2.5-VL-7B-Instruct-GGUF"}, {"title": "wanganai/qwen2.5-vl-7b-instruct · Hugging Face", "url": "https://huggingface.co/wanganai/qwen2.5-vl-7b-instruct"}, {"title": "Qwen2.5 VL 72B VRAM Needs: Why Vision Models ... - Novita", "url": "https://blogs.novita.ai/qwen2-5-vl-72b-vram/"}, {"title": "GPU System Requirements Guide for Qwen LLM Models (All Variants)", "url": "https://apxml.com/posts/gpu-system-requirements-qwen-models"}, {"title": "Fine-Tuning a Vision Language Model (Qwen2-VL-7B) | by Amit Yadav | Medium", "url": "https://medium.com/@amit25173/fine-tuning-a-vision-language-model-qwen2-vl-7b-45e78be66d30"}]

Based on the search results, here's what I found regarding the VRAM requirements for Qwen2.5-VL-7B-Instruct FP16 inference:

While the search results primarily reference Qwen 2.5-7B (the text-only version), which requires approximately 17 GB of VRAM for FP16 inference, the multimodal vision-language variant (Qwen2.5-VL-7B-Instruct) would likely require somewhat more due to its additional vision encoding components.

One source indicates that FP16 models for Qwen2.5-VL-7B variants are approximately 16GB in size, though this represents the model weights rather than total runtime VRAM. In practice, a buffer of ~20% extra VRAM is recommended for safe inference, accounting for activation memory and transient buffers.

For safe operation, a GPU with at least 24 GB VRAM (e.g., RTX 4090) would be recommended based on similar 7B-parameter models, though the actual vision-language variant may push closer to this limit.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

8. assistant (2026-05-31T04:21:33.414Z)

The searches strongly support the VRAM tier claim. Let me verify the olmoCR 2 model card directly on HuggingFace to confirm the base model and any official hardware guidance, since the claim cites the 7B Qwen2.5-VL tier specifically.

9. user (2026-05-31T04:21:39.646Z)

olmOCR-2-7B-1025 Model Details

Base Model

Yes, this model is **built on Qwen2.5-VL-7B-Instruct** and fine-tuned using the olmOCR-mix-1025 dataset.

Parameter Count

7 billion (7B) parameters (BF16 version, 8B model size)

GPU/Hardware Requirements

The page does not explicitly specify VRAM or GPU requirements. However, based on the model size (8B params in BF16), typical requirements would be:

- Minimum: ~16GB VRAM for inference
- Recommended: ~24GB+ VRAM for optimal performance

Fine-tuning and RL Training

The model underwent two stages of training:

1. **Supervised Fine-Tuning (SFT)**: Fine-tuned from Qwen2.5-VL-7B-Instruct using the [olmOCR-mix-1025](https://huggingface.co/datasets/allenai/olmOCR-mix-1025) dataset

2. **Reinforcement Learning (RL)**:

> "It has been additionally fine tuned using **GRPO RL training** to boost its performance at math equations, tables, and other tricky OCR cases."

The RL training dataset is available as [olmOCR-synthmix-1025](https://huggingface.co/datasets/allenai/olmOCR-synthmix-1025).

Note: The page recommends using the **FP8 quantized version** ([olmOCR-2-7B-1025-FP8](https://huggingface.co/allenai/olmOCR-2-7B-1025-FP8)) for practical purposes, except when fine-tuning.

10. assistant (2026-05-31T04:21:50.630Z)

I now have strong corroboration from both the primary source and the official model card. Let me assess the claim against my findings.

Verification of each sub-claim:

- "Built on Qwen2.5-VL-7B-Instruct" ? **CONFIRMED** by primary source (arXiv 2510.19817v1) and HuggingFace model card. Exact match to supporting quote.
- "7B-parameter vision-language model" ? **CONFIRMED**. Qwen2.5-VL-7B-Instruct is a 7B VLM. (Model card notes BF16 weights are ~8B param size / 8GB-class file, but the architecture is the 7B tier.)
- "Fine-tuned then RL-trained" ? **CONFIRMED**. Both sources describe SFT on olmOCR-mix-1025 followed by GRPO RL training (RLVR). Sequence is correct.
- "Anchors VRAM/hardware footprint to 7B Qwen2.5-VL tier (roughly 16-24 GB GPU class at FP16, less when quantized)" ? **SUPPORTED**. The papers/model card don't state inference VRAM explicitly, but this is a reasonable engineering inference clearly flagged as "roughly." Independent sources confirm Qwen2.5-VL-7B runs comfortably on 24GB (RTX 4090), with FP16 weights ~16GB and the text-7B needing ~17GB FP16; quantization (FP8/4-bit) reduces this. The official model card even recommends an FP8 quantized version, directly supporting "less when quantized."

The claim is well-supported by primary sources, current (Oct 2025 release, checked May 2026), and the VRAM estimate is appropriately hedged ("roughly," "class") and corroborated by multiple independent sources. No contradicting evidence found. The only minor nuance is BF16 weights being ~8B-param file size, but that does not contradict the "16-24 GB class" claim ? it reinforces it.

This claim survives adversarial scrutiny.

11. user (2026-05-31T04:21:57.439Z)

Structured output provided successfully

12. assistant (2026-05-31T04:22:01.813Z)

I've completed the verification. The claim survives adversarial scrutiny (refuted=false, high confidence).

The architectural core of the claim is directly and exactly supported by the primary source quote, and additionally confirmed by the official HuggingFace model card ? both state olmOCR 2 is built on Qwen2.5-VL-7B-Instruct (7B VLM) with SFT followed by GRPO RL training. The VRAM estimate ("roughly 16-24 GB class at FP16, less when quantized") is appropriately hedged and corroborated by multiple independent sources, including the model card's own recommendation of an FP8 quantized variant. No contradicting evidence was found.